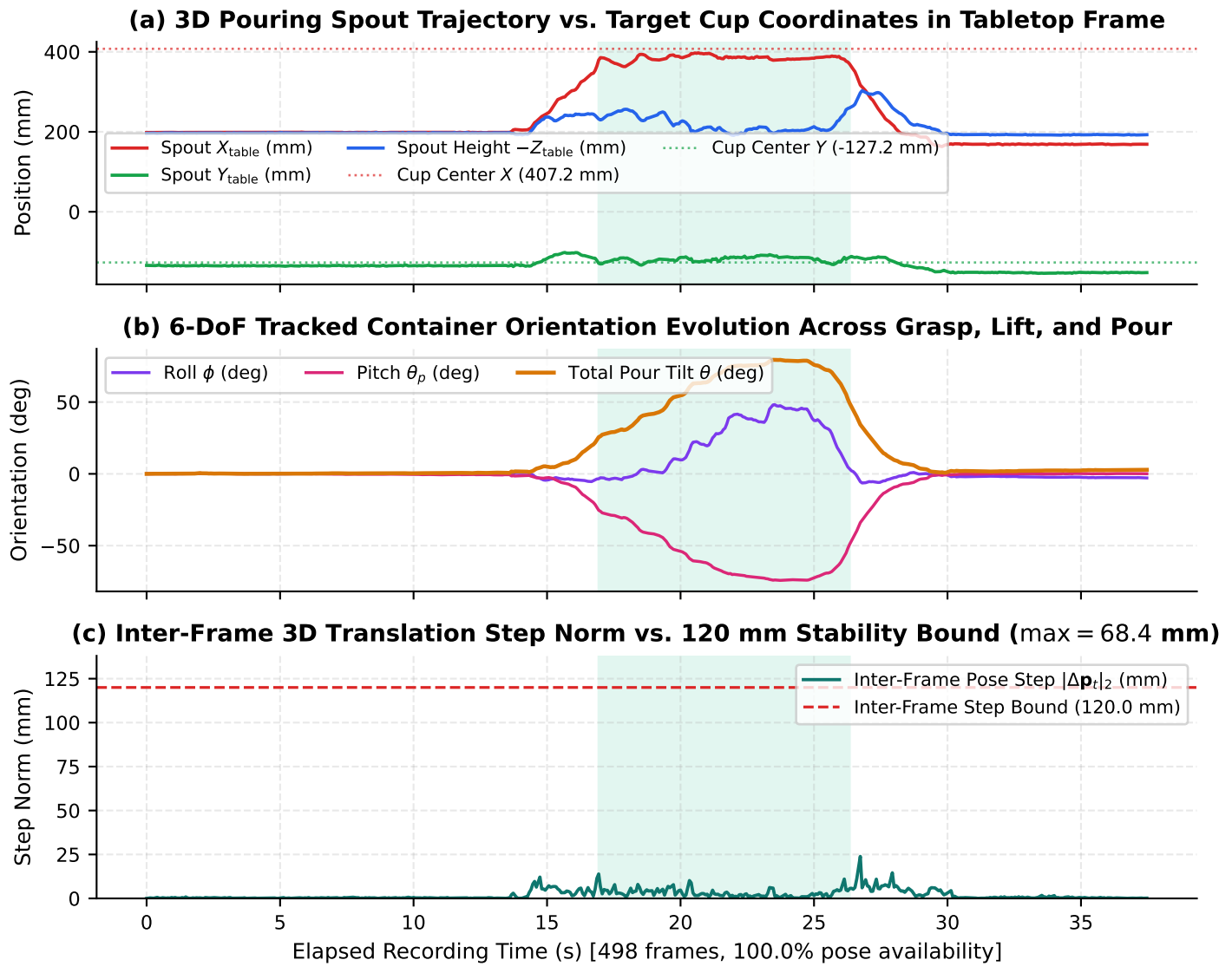

(a) 3D Pouring Spout Trajectory vs. Target Cup Coordinates in Tabletop Frame
Position (mm)
400
200
0
Spout X_{table} (mm)
Spout Y_{table} (mm)
Spout Height $-Z_{table}$ (mm)
Cup Center X (407.2 mm)
Cup Center Y (-127.2 mm)
(b) 6-DoF Tracked Container Orientation Evolution Across Grasp, Lift, and Pour
Orientation (deg)
50
0
−50
Roll ϕ (deg)
Pitch θ_p (deg)
Total Pour Tilt θ (deg)
(c) Inter-Frame 3D Translation Step Norm vs. 120 mm Stability Bound (max = 68.4 mm)
Step Norm (mm)
125
100
75
50
25
0
Inter-Frame Pose Step $|\Delta \mathbf{p}_t|_2$ (mm)
Inter-Frame Step Bound (120.0 mm)
0
5
10
15
20
25
30
35
Elapsed Recording Time (s) [498 frames, 100.0% pose availability]